

Syntax: Introduction

Till now, we have discussed:

Language modeling — understanding ordering of words

POS tagging — understanding roles of words

Now we will discuss more complex notions, such as grammatical relations among words

What is Syntax?

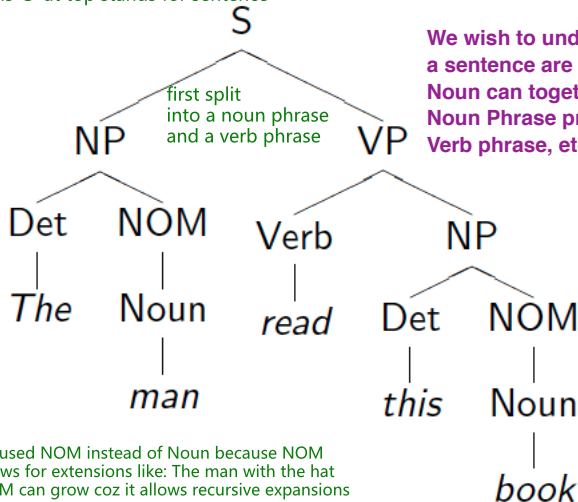
- Refers to the way words are arranged together, and the relationship between them.

Syntax Tree: Example

This 'S' at top stands for sentence

Sentence: The man read this book

We wish to understand how different words in a sentence are grouped, e.g., a Det and a Noun can together give a Noun Phrase, a Noun Phrase preceded by a verb can give a Verb phrase, etc.



we used NOM instead of Noun because NOM allows for extensions like: The man with the hat
NOM can grow coz it allows recursive expansions

Our final goal:
How to construct a syntax tree for a given sentence?

Defining the notions: Constituency

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Part of Speech - "Substitution Test"

The {sad, intelligent, green, fat, ...} one is in the corner.

Different words having the same POS (here, adjective) can be used as a substitute for each other.

Constituency: Noun Phrase

- *Kermit the frog*
- *they*
- *December twenty-sixth*
- *the reason he is running for president*

Similarly, different word-sequences (phrases) can be used as a substitute for each other.

E.g., noun phrases

These phrases are very different from each other, But syntactically, they behave the same way because all are noun phrases

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Kermit the frog comes on stage

They come to Massachusetts every summer

December twenty-sixth comes after Christmas

The reason he is running for president comes out only now.

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*The comes out... *is comes out... *for comes out...

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*The comes out... *is comes out... *for comes out...

The underlined parts are very different, but they all appear before , comes..

They all occupy the same position in the sentence, which is the subject position

Can be placed in a number of different locations

The Subject position requires a full constituent, not just any word.

Constituent = Prepositional phrase: On December twenty-sixth

On December twenty-sixth I'd like to fly to Florida.

I'd like to fly on December twenty-sixth to Florida.

I'd like to fly to Florida on December twenty-sixth.

But not split apart

*On December I'd like to fly twenty-sixth to Florida.

*On I'd like to fly December twenty-sixth to Florida.

The entire prepositional phrase can move as a block

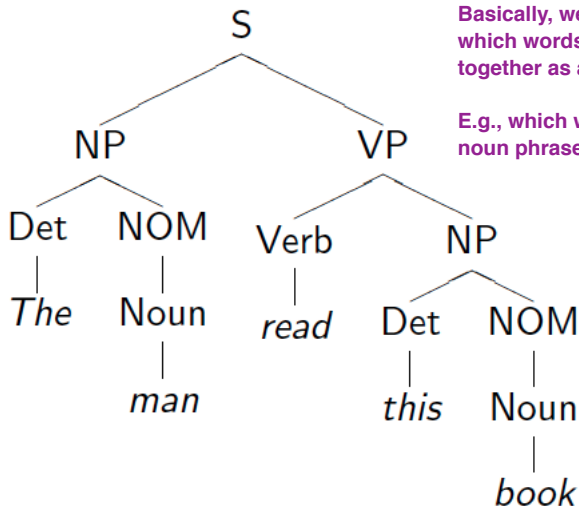
But if u split the prepositional phrase into parts and u move only part of it, then Grammar rejects it.

So constituent moves together as a single block, you cannot break them into parts

Modeling Constituency: what tool do we need?

Basically, we want tools to understand which words (in a given text) come together as a unit.

E.g., which words together make up a noun phrase?



Modeling Constituency

Context-free grammar

The most common way of modeling constituency

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These rules express the ways in which the symbols of the language can be grouped and ordered together

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Example

Noun phrase can be composed of either a ProperNoun or a determiner (Det) followed by a Nominal; a Nominal can be more than one nouns

$NP \rightarrow \text{Det Nominal}$

$NP \rightarrow \text{ProperNoun}$

$\text{Nominal} \rightarrow \text{Noun} \mid \text{Noun Nominal}$

CFG: $G = (T, N, S, R)$

- T : set of terminals
- N : set of non-terminals
 - ▶ For NLP, we distinguish out a set $P \subset N$ of pre-terminals, which always rewrite as terminals
- S : start symbol
- R : Rules/productions of the form $X \rightarrow \gamma$, $X \in N$ and $\gamma \in (T \cup N)^*$

Terminals and pre-terminals

Terminals mainly correspond to words in the language while pre-terminals mainly correspond to POS categories

Example

$NP \rightarrow \text{Det Nominal}$

$NP \rightarrow \text{ProperNoun}$

$\text{Nominal} \rightarrow \text{Noun} \mid \text{Noun Nominal}$

“Noun”, “ProperNoun”, “Det” are pre-terminals.

But we cannot generate any string with these rules, since there is no terminal.

Example

$NP \rightarrow \text{Det Nominal}$

$NP \rightarrow \text{ProperNoun}$

$Nominal \rightarrow \text{Noun} \mid \text{Noun Nominal}$

Now, these can be combined with other rules, that express facts about a lexicon.

$Det \rightarrow a$ These rules now connect the abstract grammar symbols to real words in the lexicon.

$Det \rightarrow the$

$Noun \rightarrow flight$

A noun phrase can be expanded into a determiner followed by a Nominal
If you see an NP node in a tree,
you are allowed to replace it with: DET Nominal

Terminals: “a”, “the”, “flight”

Pre-terminals: “Det”, “Noun”, “ProperNoun”

Non-terminals: “NP”, “Nominal”

CFG as a generator

NP → Det Nominal

NP → ProperNoun

Nominal → Noun | Noun Nominal

Det → a

Det → the

Noun → flight

CFG as a generator

$NP \rightarrow \text{Det Nominal}$

$NP \rightarrow \text{ProperNoun}$

$\text{Nominal} \rightarrow \text{Noun} \mid \text{Noun Nominal}$

$\text{Det} \rightarrow \text{a}$

$\text{Det} \rightarrow \text{the}$

$\text{Noun} \rightarrow \text{flight}$

Generating 'a flight':

CFG as a generator

$NP \rightarrow \text{Det Nominal}$

$NP \rightarrow \text{ProperNoun}$

$\text{Nominal} \rightarrow \text{Noun} \mid \text{Noun Nominal}$

$\text{Det} \rightarrow \text{a}$

$\text{Det} \rightarrow \text{the}$

$\text{Noun} \rightarrow \text{flight}$

Generating 'a flight':

$NP \rightarrow \text{Det Nominal}$

$\rightarrow \text{Det Noun} \rightarrow \text{a Noun} \rightarrow \text{a flight}$

CFG as a generator

$NP \rightarrow \text{Det Nominal}$

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Generating 'a flight':

$NP \rightarrow \text{Det Nominal}$

$\rightarrow \text{Det Noun} \rightarrow \text{a Noun} \rightarrow \text{a flight}$

In generation step, we start with a non-terminal

Here we want to generate a Noun Phrase so we start with :

NP - this is just an abstract symbol

Apply Production Rule :

we look at rules which can expand the NP.

We have two options but since "A Flight" has DETERMINER "a"

we choose first rule, $NP \rightarrow \text{Det Nominal}$

Next, we expand $\text{DET} \rightarrow \text{a}$ and then string becomes a Nominal

Then we expand Nominal

- Thus a CFG can be used to randomly generate a series of strings

NP

$\rightarrow \text{Det Nominal}$

$\rightarrow \text{a Nominal}$

$\rightarrow \text{a Noun}$

$\rightarrow \text{a flight}$

CFGs and Grammaticality

A CFG defines a formal language = set of all sentences (string of words) that can be derived by the grammar

- Sentences in this set are said to be **grammatical**
- Sentences outside this set are said to be **ungrammatical**

If you start from the start symbol (usually S) and repeatedly apply the grammar rules until only words remain, every string you can produce belongs to that language

If the grammar can generate a sentence using its rules, that sentence is considered grammatical according to that grammar.

CFGs and Recursion

A CFG can contain recursive rules.

E.g., the rule for a Prepositional Phrase (PP)

Recursive Definition

- $PP \rightarrow \text{Prep } NP$
- $NP \rightarrow \text{Noun } PP$

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- $PP \rightarrow \text{Prep NP}$
- $NP \rightarrow \text{Noun PP}$

Example Sentence

[_SThe mailman ate his [_{NP} lunch [_{PP} with his friend [_{PP} from the cleaning staff [_{PP} of the building [_{PP} at the intersection [_{PP} on the north end [_{PP} of town]]]]]]].

start of a NP

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